CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

TVEL FINANCIAL NETWORK ARCHITECTURE ASSESSMENT

Rosatom Commercial Fuel Cycle — Banking Structure, SPV Evasion Patterns, CNY Settlement, Isotope JSC Financial Exposure

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

KEY JUDGMENTS

KJ 1

TVEL JSC's April 2024 OFAC SDN designation has been materially diluted by three factors: legacy contract carve-outs for allied reactor operators; migration to CNY-denominated settlement for Chinese customers; and SPV structures that route transactions around U.S. correspondent banking requirements.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

Isotope JSC — a Rosatom subsidiary distributing proliferation-relevant isotopes including Cf-252 — is not OFAC-designated as of March 2026 and maintains active Western commercial relationships. Isotope JSC designation is the highest-priority low-collateral, high-accessibility financial enforcement action against the Russian fuel cycle. It is not the highest system-disruption action (that is UECC, which carries catastrophic collateral), but it is the action with the best ratio of enforcement feasibility to nonproliferation impact.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 3

Post-2022 migration of TVEL-CNNC contracts to CNY settlement through Chinese state banks (ICBC, Bank of China) has reduced U.S. financial enforcement leverage over the Russian-Chinese nuclear fuel relationship to near zero under current instruments.

LIKELY — MODERATE-HIGH CONFIDENCE

KJ 4

TVEL's evasion architecture in Geneva, Dubai, and Istanbul — SPV structures exploiting OFAC's 50% ownership rule — is assessed as the primary mechanism allowing sophisticated actors to continue transacting with TVEL without triggering blocking obligations. Specific SPV entities are not identified in open-source evidence. Network resolution requires Swiss Handelsregister and UAE commercial registry data beyond current OSINT scope. This assessment describes an evasion architecture, not a mapped network.

LIKELY — MODERATE CONFIDENCE — ARCHITECTURALLY INFERRED

I. TVEL CORPORATE STRUCTURE AND REVENUE STREAMS

ATTRIBUTION	TVEL JSC was designated by OFAC under EO 13382 on April 19, 2024 [TIER 1 — OFAC SDN List]. Rosatom's 2023 Annual Report identifies TVEL JSC as the commercial parent of the Russian enrichment and fabrication network. The Congressional Research Service's 2023 analysis (CRS R47742) estimated TVEL's pre-sanctions annual enrichment revenue at $2–3 billion [TIER 1 — CRS R47742, 2023].
JUDGMENT	Official reporting supports a high-confidence judgment that TVEL's April 2024 SDN designation is the most significant enforcement action against the Russian fuel cycle to date, but that its practical effect has been substantially diluted by the three evasion factors identified above.
LIMITS	The public record does not establish the current total value of TVEL transactions still clearing through Western financial infrastructure, the specific correspondent banks currently active, or the precise contract terms of OFAC General License wind-down provisions for fuel cycle transactions.

A. Revenue Streams by Category

REVENUE TYPE	ESTIMATED VOLUME	CLEARING MECHANISM
SWU (enrichment services)	~$2–3B annually pre-2022 [CRS R47742]	USD/EUR correspondent banking — migrating to CNY/bilateral post-2022
Fuel assembly exports	~$1–2B annually	USD/EUR legacy contracts; CNY for China post-2022
Conversion services	~$300–500M	Bundled in fuel supply contracts
Isotope sales (Isotope JSC)	~$200–400M	USD/EUR — partially via Western pharma customers — NOT DESIGNATED
Uranium trading (TENEX)	Variable	Multi-currency; partially via Geneva trading entity

II. CORRESPONDENT BANKING ARCHITECTURE

ATTRIBUTION	OFAC General License 8 and its renewals granted wind-down periods for energy sector transactions with designated Russian entities. State Department reporting and Congressional Research Service analysis document that nuclear fuel-specific carve-outs allowed EU and Asian reactor operators to continue receiving fuel under pre-existing contracts [TIER 1 — OFAC GL8; CRS R47742].
JUDGMENT	Available reporting supports a moderate-confidence judgment that legacy contract carve-out wind-down periods may still sustain residual Western banking activity for TVEL SWU revenue. The public record does not permit a firm conclusion on whether all applicable wind-down periods have expired as of March 2026.
LIMITS	The public record does not establish which specific correspondent banks are currently clearing residual TVEL SWU revenue, which EU and Asian operators are still active under carve-out arrangements, or the total remaining Western financial exposure to TVEL transactions.

IMPLEMENTATION FRICTION Post-2022 migration to EUR, CNY, and bilateral currency arrangements has substantially reduced USD-denominated TVEL clearing through U.S. correspondent banks. The residual enforcement leverage is in EUR-denominated transactions through European banks and in the SPV network maintaining commercial relationships with Western counterparties.

III. SPV EVASION ARCHITECTURE — HYPOTHESIS ASSESSMENT

ATTRIBUTION	OFAC's 50% ownership rule provides that entities owned 50% or more by an SDN-listed entity are constructively blocked even without explicit listing [TIER 1 — OFAC Guidance on Entities Owned by Blocked Persons]. This rule creates an incentive to structure entities below the 50% threshold. This section assesses the architecture of likely evasion structures — it does not map a confirmed network, as specific SPV entities are not identified in open-source evidence.
JUDGMENT	Available reporting supports a moderate-confidence judgment that TVEL's SPV network in Geneva (uranium trading/TENEX), Dubai (post-2022 energy transactions), and Istanbul (emerging clearing jurisdiction) allows transactions to route around TVEL's SDN designation. Systematically identifying and designating this network would close the most significant gap in TVEL SDN enforcement.
LIMITS	The public record does not establish the specific SPV entities in these jurisdictions, their ownership structures relative to the 50% rule, or the transaction volumes routing through each jurisdiction. Allied financial intelligence from Swiss FINMA, UAE Central Bank, and BDDK Turkey warrants attention to map this network.

IMPLEMENTATION FRICTION OSINT ceiling note: This section describes an inferred evasion architecture based on known OFAC rule structures and documented Russian financial adaptation patterns. No specific SPV entity names, ownership chains, or transaction flows are documented in open-source evidence. The assessment

is labeled ARCHITECTURALLY INFERRED. Swiss FINMA filings, UAE commercial registrations, and Turkish BDDK data would be required to resolve this hypothesis into a mapped network.

IV. ISOTOPE JSC FINANCIAL NETWORK

ATTRIBUTION	Isotope JSC is identified in Rosatom commercial documentation as the subsidiary handling isotope distribution from NIIAR, ECP, and NCCP [TIER 2 — Rosatom commercial documentation; TIER 3 — Nuclear Engineering International]. BIS ECCN 1C233 controls lithium-6 and ECCN 1C236 controls Cf-252 as nuclear-related materials.
JUDGMENT	Available reporting supports a high-confidence judgment that Isotope JSC's Western pharmaceutical and industrial customer relationships represent an active, undesignated Western financial flow from Russian nuclear facility operations. This is the most significant single gap in current TVEL-related enforcement coverage.
LIMITS	The public record does not establish the specific Western pharmaceutical companies purchasing medical isotopes from Isotope JSC, the contract values, the clearing banks, or the end-use verification procedures for weapons-relevant isotopes including Cf-252 and lithium-6.

V. CNY SETTLEMENT ARCHITECTURE

ATTRIBUTION	Congressional Research Service analysis documents that post-2022 Russia-China energy transactions have substantially migrated to CNY settlement through Chinese state banks [TIER 1 — CRS R47742, 2023; TIER 1 — Treasury Department Russia sanctions reporting].
JUDGMENT	Official reporting supports a moderate-high-confidence judgment that TVEL-CNNC contract settlement has migrated to CNY clearing through ICBC, Bank of China, and CITIC, placing these transactions outside U.S. financial monitoring infrastructure. Restoring leverage requires either secondary sanctions against Chinese clearing banks, a multilateral NSG instrument, or supply substitution.
LIMITS	The public record does not establish the specific CNY clearing architecture for TVEL-CNNC transactions, the total CNY-denominated SWU revenue volume post-2022, or whether any residual USD touchpoints exist in the TVEL-CNNC transaction chain.

VI. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Pursuit of Isotope JSC SDN designation immediately — the highest-priority low-collateral, high-accessibility financial enforcement action in the index. Isotope JSC carries no reactor safety collateral (D3=5), unilateral enforcement capability (D4=5), and a documented Western financial

nexus through the Nordion-Lantheus-Wells Fargo chain (FPA-01-v3). It is not the highest-leverage node (UECC) but it is the most immediately executable with the best enforcement-to-impact ratio.

2. Financial intelligence mapping of and designate TVEL's SPV network in Geneva, Dubai, and Istanbul through allied financial intelligence coordination with Swiss FINMA, UAE Central Bank, and BDDK.

3. Audit of legacy contract carve-out expiry for EU and Asian reactor operators — confirm whether TVEL clearing through Western banks has ceased or continues under extended wind-down arrangements.

4. Designation of TENEX JSC (uranium trading) to close the uranium trading financial channel separate from TVEL's direct fuel sales.

5. Development of a secondary sanctions framework assessment for Chinese banks clearing TVEL-CNNC CNY transactions — establish legal predicate and cost-benefit analysis before execution.

6. Pursuit of coordinated NSG-level instruments to create non-financial restrictions on TVEL contracts with member state reactor operators.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

SANCTIONS EVASION INFRASTRUCTURE

Russian Nuclear Fuel Cycle — Procurement Networks, Maritime Evasion, Financial Intermediaries

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

KEY JUDGMENTS

KJ 1

UAE free trade zone front companies — primarily in Jebel Ali and Dubai Multi Commodities Centre — are assessed as a leading conduit for Western dual-use technology procurement for Russian nuclear facilities post-2022. 'Leading conduit' reflects the volume and frequency of UAE-origin shipment interdictions and Entity List additions documented in BIS enforcement releases; it does not rest on a comparative share estimate for which open-source volume data is insufficient. BIS Entity List additions have not kept pace with new front company incorporation.

LIKELY — MODERATE CONFIDENCE

KJ 2

Rosatom's Akkuyu NPP construction contract in Turkey creates a documented conflict of interest: Turkish authorities have commercial and diplomatic incentive to maintain trade relationships with Rosatom entities including those involved in dual-use procurement.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 3

Post-2022 Russia-DPRK maritime corridor expansion uses the Vostochny/Nakhodka complex as the primary departure point. AIS manipulation and flag-hopping are documented evasion techniques in this corridor [TIER 1 — UN Panel of Experts S/2024/215].

ALMOST CERTAINLY — HIGH CONFIDENCE

I. DUAL-USE PROCUREMENT NETWORKS

A. UAE Procurement Hub

ATTRIBUTION

BIS Entity List enforcement actions and Treasury OFAC enforcement releases document UAE-based front companies as re-export conduits for Western technology to Russia post-2022 [TIER 1 — BIS Entity List additions 2022–2024; OFAC

	enforcement releases]. UAE free trade zones (Jebel Ali, Dubai Multi Commodities Centre) are specifically identified in enforcement actions.
JUDGMENT	Official reporting supports a moderate-confidence judgment that UAE free trade zones remain the leading third-country hub for Russian nuclear facility dual-use procurement. The pace of BIS Entity List additions is insufficient to disrupt the network given the rate of new front company incorporation.
LIMITS	The public record does not establish the specific entities currently procuring nuclear facility dual-use equipment through UAE channels, the specific equipment categories, or the total procurement volume. BIS enforcement actions represent a subset of actual activity.

B. Turkey — Akkuyu Conflict of Interest

ATTRIBUTION	Rosatom's Akkuyu NPP construction in Turkey is documented in IAEA country reports and Turkish Nuclear Regulatory Authority filings as an active government-to-government construction project [TIER 1 — IAEA CNPP Turkey 2024; TIER 2 — Turkish TAEK reports].
JUDGMENT	Official reporting supports a high-confidence judgment that Rosatom's Akkuyu construction creates a structural conflict of interest for Turkish export control enforcement against Rosatom-affiliated entities. Turkish intermediaries purchasing Western technology for Russian nuclear facilities face limited domestic enforcement pressure for this reason.
LIMITS	The public record does not establish the specific Turkish entities currently serving as procurement intermediaries for Russian nuclear facilities, or the categories of equipment being transited through Turkey.

C. China and Central Asia

Chinese entities provide both direct procurement and technology substitution — Chinese-manufactured equivalents for Western components subject to export controls. Technology substitution is increasingly significant: China has domestic manufacturing capability for many precision components required for centrifuge maintenance. Chinese customs data is not publicly available and Chinese intermediaries are not consistently identifiable in Western compliance databases. Kazakhstan warrants specific attention: as both a uranium producer with Rosatom involvement (Kazatomprom/Uranium One) and a transit state, it creates a Western-accessible financial interface in the Russian-adjacent supply chain that is currently underutilized for enforcement purposes.

II. MARITIME EVASION INFRASTRUCTURE

ATTRIBUTION	UN Panel of Experts reports to the UNSC 1718 Committee document AIS manipulation, flag-hopping, and STS transfer as documented evasion techniques in the Russia-DPRK maritime corridor [TIER 1 — UN Panel of Experts S/2024/215;

	S/2023/171]. Post-2022 Russian fleet flag-hopping to Palau, Cameroon, Gabon, Togo, and Comoros registries is specifically identified.
JUDGMENT	Official UN reporting supports a high-confidence judgment that the Russia-DPRK maritime corridor operates with systematic AIS manipulation and flag-hopping as standard practice. These techniques are documented, not inferred.
LIMITS	The public record does not establish the specific vessels currently operating the Russia-DPRK corridor out of Vostochny, or whether any nuclear-relevant material has been transported in this corridor. UN Panel reporting covers a subset of active vessels.

III. FINANCIAL INTERMEDIARY INFRASTRUCTURE

ATTRIBUTION	Treasury Department Russia sanctions reporting and Congressional Research Service analysis document Turkish state banks (Ziraat Bank, Vakifbank), UAE domestic banks, and Chinese state banks (ICBC, Bank of China, CITIC) as major financial intermediary channels for Russian sanctions-evasion transactions post-2022 [TIER 1 — Treasury Department Russia sanctions reports 2023–2024; CRS R47742].
JUDGMENT	Official reporting supports a moderate-high-confidence judgment that the combination of Turkish state bank correspondent relationships and CIPS-based CNY clearing through Chinese state banks constitutes the leading surviving financial infrastructure for Russian nuclear fuel cycle transactions post-TVEL SDN designation.
LIMITS	The public record does not establish the specific transaction volumes routing through each intermediary channel, or the extent to which cryptocurrency is used for nuclear fuel cycle-related transactions versus other Russian sanctions-evasion activity.

IV. POLICY OPTIONS AND ANALYTICAL RECOMMENDATIONS

1. Acceleration of Entity List additions for UAE free trade zone entities identified as Russian nuclear dual-use procurement fronts.

2. Engagement through UAE, Turkish, and Indian authorities through bilateral export control cooperation frameworks to strengthen domestic enforcement against re-export violations.

3. An OFAC action to target Turkish state bank correspondent relationships — specifically Ziraat Bank and Vakifbank USD correspondent accounts — as the leverage point for Turkish financial intermediary enforcement.

4. Implementation of vessel port-of-call restrictions in allied Pacific jurisdictions for all vessels with documented AIS manipulation or flag-hopping histories in the Russia-DPRK corridor.

5. Development of a CIPS-nexus secondary sanctions framework: establish legal basis for targeting non-U.S. entities that use CIPS to clear transactions for OFAC-designated Russian nuclear entities.

6. State and A Treasury action to coordinate with Five Eyes partners on Kazatomprom-Uranium One financial flows as a Western-accessible enforcement interface in the Russian-adjacent uranium supply chain.

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WP-2026 SERIES

ENFORCEMENT SEQUENCING ANALYSIS

SENI 44-Node Index — Priority Order, Coalition Requirements, CCRP Application

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

KEY JUDGMENTS

KJ 1

Eight nodes in the SENI index are immediately actionable by the U.S. without allied coordination and without triggering reactor safety or multilateral constraints. Isotope JSC designation and NIIAR Cf-252 ECCN controls are the highest-priority immediate actions.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

Full UECC designation — the highest-leverage single action in the index — is preconditioned on allied enrichment supply buildout reaching ≥85% of UECC displaced volume. This precondition is not met as of March 2026.

LIKELY — MODERATE-HIGH CONFIDENCE

KJ 3

The Coalition Conflict Resolution Protocol (CCRP) from WP-2026-SHIELD-01 identifies three primary coalition fault lines in the nuclear enforcement context: EU reactor operator friction, China friction at Krasnokamensk and Angarsk, and IAEA institutional friction. Each requires a distinct resolution pathway.

ALMOST CERTAINLY — HIGH CONFIDENCE

I. ANALYTICAL FRAMEWORK

This document applies the Coalition Conflict Resolution Protocol (CCRP) from WP-2026-SHIELD-01 to the full 44-node SENI enforcement index. The CCRP framework distinguishes between actions executable unilaterally by the U.S., actions requiring allied coordination, and actions constrained by multilateral legal architecture. Sequencing recommendations reflect three criteria: enforcement leverage (expected disruption to Russian nuclear fuel cycle operations); collateral cost (reactor safety risk, allied diplomatic friction, market disruption); and precondition dependency (actions that warrants precede others to be effective).

II. NODE CLASSIFICATION BY ACTION TIER

Tier 1 — Unilateral U.S. Action Available Now

NODE	ACTION	AUTHORITY	CONSTRAINT
33 — NCCP	Isotope JSC SDN + BIS Entity List	EO 13382 / ECCN 1C233	None — independent Li-6 predicate
30 — ECP	Isotope JSC BIS Entity List	EAR ECCN 1C233	Isotope JSC separate action required
43 — NIIAR	Cf-252 BIS ECCN 1C236 controls	EAR	Immediately actionable
44 — Kurchatov	BIS Entity List + visa controls	EAR / Admin	Political cost moderate
34 — Chepetsk	Explicit SDN listing (TVEL sub)	EO 13382	Verify OFAC coverage first
32 — Elektrostal	Explicit SDN listing (TVEL sub)	EO 13382	Verify OFAC coverage first
42 — Zheleznogorsk	SDN + RT-2 procurement targeting	EO 13382	No Western leverage on Pu ops
41 — Mayak (commercial)	Commercial reprocessing SDN	EO 13382	Weapons Pu ops unaffected — PMDA track separate

Tier 2 — Requires Allied Coordination

NODE	ACTION	PRECONDITION	COALITION
28 — UECC	Full SDN designation	Urenco+Orano+Centrus ≥85% UECC displacement	Five Eyes + EU + Japan
29 — SCC	SDN designation	IAEA safeguards consultation complete	Five Eyes + IAEA
36 — Vostochny	FESCO + port-of-call restrictions	Allied Pacific port coordination	Japan, South Korea, Australia
35 — Zabaykalsk	Logistics intermediary designation	Rusatom Service mapping complete	Five Eyes logistics intel

37 — Tayshet	RZhD designation assessment	Cost-benefit analysis complete	EU + allied economic coord

Tier 3 — Constrained / Long-Lead

NODE	CONSTRAINT TYPE	RESOLUTION PATH
27 — Krasnokamensk	TVEL nuclear safety / Chinese reactor ops	Define fuel stockpile threshold; Chinese utility transition planning
31 — Angarsk (AECC)	IUEC multilateral architecture	Surgical OFAC carve-out OR IUEC member state transition to alt supply
38 — Sarov	No financial exposure; arms control domain	BIS Entity List for procurement; arms control monitoring restoration
39 — Trekhgorny	No financial exposure	Export controls on precision manufacturing procurement
40 — Lesnoy	No financial exposure	Export controls on uranium metallurgy equipment
41 — Mayak (Pu ops)	No financial exposure; PMDA domain	PMDA restoration or successor mechanism

III. CCRP APPLICATION — COALITION CONFLICT MATRIX

ATTRIBUTION	WP-2026-SHIELD-01 establishes the Coalition Conflict Resolution Protocol (CCRP) as the governing framework for managing coalition friction in coordinated enforcement actions. The CCRP identifies three categories of conflict: structural (market or supply constraints), diplomatic (allied government friction), and institutional (multilateral body friction).
JUDGMENT	Applying the CCRP to the SENI index supports a high-confidence judgment that the three primary coalition fault lines — EU reactor operator friction, China friction at Krasnokamensk and Angarsk, and IAEA institutional friction — each require distinct resolution pathways. Applying a single enforcement strategy across all nodes without accounting for these fault lines would generate coalition fracture.
LIMITS	The public record does not establish the specific positions of EU member states, China, or IAEA on the specific enforcement actions recommended in this index. Coalition positions would need to be assessed through diplomatic engagement before execution of any Tier 2 action.

Resource Constraint Assumption

The sequencing logic presented in this document assumes that enforcement actions are not simultaneously capacity-constrained. In practice, OFAC designation package development, State Department coordination, and BIS Entity List additions each require finite analyst-hours and political capital. The following priority ordering addresses the 'if only one action is possible' constraint explicitly.

RANK	ACTION	RATIONALE FOR PRIORITY
1	Isotope JSC SDN designation	Highest enforcement-to-impact ratio. Package predicate established (FPA-01-v3). No allied coordination required. No reactor safety risk. Immediately executable.
2	BIS ECCN 1C236 enhanced end-use (Cf-252)	Available without OFAC coordination. Closes the most specific WMD-relevant material gap. Reinforces Isotope JSC designation predicate.
3	ChMP post-GL 115A secondary sanctions notification	Date-certain trigger (June 30, 2025). Requires coordination with EU allies. Secondary preparation ongoing.
4	UECC designation package preparation (not execution)	High future leverage. Preparation now reduces lead time when precondition is met. Does not require current execution.

EU reactor operator friction is resolved by supply precondition sequencing — execute UECC designation only after Urenco/Orano/Centrus capacity milestone is confirmed. China friction at Krasnokamensk and Angarsk is managed through surgical designation carve-outs and threshold conditions rather than blanket enforcement. IAEA institutional friction is managed by framing enforcement actions as complementary to enhanced safeguards — engaging IAEA proactively on safeguards gaps at target facilities before executing designation actions.

IV. PRIORITY EXECUTION SEQUENCE

1. IMMEDIATE (no preconditions): Isotope JSC SDN designation; NIIAR Cf-252 BIS ECCN 1C236 controls; Kurchatov BIS Entity List + visa controls; Chepetsk and Elektrostal explicit SDN listing confirmation with OFAC.
2. SHORT-TERM (30–90 days): TVEL SPV network mapping and designation in Geneva, Dubai, Istanbul; Zheleznogorsk SDN paired with RT-2 procurement targeting; Mayak commercial operations SDN.
3. MEDIUM-TERM (90–180 days, precondition-dependent): FESCO and Vostochny Port designation coordinated with Pacific allies; Rusatom Service logistics intermediary designation; Zabaykalsk corridor freight forwarder mapping complete.
4. LONG-TERM (supply-precondition-dependent): UECC full SDN designation (Urenco/Orano/Centrus capacity milestone ≥85%); SCC SDN (IAEA consultation complete); Krasnokamensk threshold condition defined and met.

5. ARMS CONTROL TRACK (parallel, non-sanctions): PMDA restoration or successor mechanism for Mayak/Zheleznogorsk Pu accountability; IAEA enhanced safeguards at Seversk, Angarsk, Zheleznogorsk; ADE-2 Pu accountability gap resolution.

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CONFIDENCE FRAMEWORK

SENI Index — Evidentiary Basis, Source Quality, and Analytical Uncertainty

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KEY JUDGMENTS

KJ 1

Four assertions in the current SENI index carry LOW CONFIDENCE — ANALYTICALLY INFERRED status and require either corroboration from Tier 1 or credible Tier 2 sources, or explicit downgrading before submission to SSRN or Georgetown SSP.

ALMOST CERTAINLY — HIGH CONFIDENCE

KJ 2

Complete absence of academic literature engagement across SENI node profiles is the single most critical gap for Georgetown SSP submission, consistent with the March 2026 WP-2026 portfolio review finding.

ALMOST CERTAINLY — HIGH CONFIDENCE

I. CONFIDENCE TIER DEFINITIONS

Probability and confidence are separate analytical dimensions, applied separately to every analytic judgment per WP-2026 Style Guide v3.0. Probability describes the likelihood that the judgment is correct. Confidence describes the quality and completeness of supporting evidence.

TIER	LABEL	DEFINITION	EXAMPLE SOURCES
C1 / HIGH	HIGH CONFIDENCE	Verified in multiple independent open-source documents; consistent with IAEA safeguards declarations or official government disclosures	IAEA reports, OFAC SDN List, Rosatom annual reports, official government filings, peer-reviewed technical literature, NRC/DOE public disclosures, CRS reports
C2 / MOD-	MODERATE-HIGH /	Reported by credible open-source analysis; consistent with known facility	NTI profiles, RAND, CSIS, Carnegie, Brookings, WNA

HIGH	MODERATE	characteristics; limited independent corroboration	reports, Foreign Affairs, TNSR, congressional testimony with named attribution
C3 / LOW-MOD	LOW-MODERATE CONFIDENCE	Analytically inferred from facility type, geographic position, or known Russian nuclear complex structure; not directly reported in open sources	Author inference from open-source facility data, geographic analysis, comparative analysis with analogous facilities — warrants be labeled ANALYTICALLY INFERRED
C4 / LOW	LOW CONFIDENCE	Reported in limited sources of variable reliability; single-source or secondary reporting; requires verification before use as enforcement predicate	Single-source reporting, secondary sources, unverified claims — warrants be labeled ANALYTICALLY INFERRED; cannot serve as published enforcement predicate

II. NODE-BY-NODE CONFIDENCE ASSESSMENT

NODE	CONFIDENCE	KEY UNCERTAINTY / LIMITS
27 — Krasnokamensk	C1 / HIGH	Facility identity and production role C1. TVEL fuel supply constraints C1–C2. Specific tonnage figures C2. Threshold condition for action C3.
28 — Novouralsk (UECC)	C1 / HIGH	48% Russian capacity C1 (Rosatom). 20%+ global figure C1–C2 (IAEA/WNA). Allied supply capacity precondition C2.
29 — Seversk (SCC)	C1–C2	Civilian operations C1. Weapons history C1. Current dual-use infrastructure status C2–C3.
30 — Zelenogorsk (ECP)	C1–C2	Enrichment role C1. Isotope production details C2. Dual-use isotope concerns C3.
31 — Angarsk (AECC)	C1	IUEC structure and IAEA involvement C1 (publicly documented treaty framework).
32 — Elektrostal	C1	Fuel fabrication role and reactor types C1. Export customer

		status post-2022 C2–C3.
33 — NCCP Novosibirsk	C1–C2	Fuel fabrication C1. Li-6 production C1–C2. DPRK procurement connection C3–C4 — ANALYTICALLY INFERRED.
34 — Chepetsk (ChMP)	C1	Conversion and zirconium roles C1. Uranium metal production details C2.
35 — Zabaykalsk	C1	Rail crossing identity and gauge-change function C1. Nuclear manifest monitoring capability C3 — ANALYTICALLY INFERRED.
36 — Vostochny Port	C1–C2	Port identity C1. Russia-DPRK corridor use C1 (UN Panel reporting). Nuclear material routing C2–C3.
37 — Tayshet Junction	C1	Rail junction identity C1. BAM routing for anomalous nuclear shipments C3 — ANALYTICALLY INFERRED.
38 — Sarov (VNIIEF)	C1	Facility identity and design role C1. Technology transfer to DPRK C4 — ANALYTICALLY INFERRED.
39 — Trekhgorny	C1–C2	Facility identity and assembly role C1–C2. Specific capabilities C2–C3.
40 — Lesnoy	C1–C2	Facility identity C1. HEU metal production C2. Current operational status C2–C3.
41 — Mayak	C1	Facility identity and reprocessing role C1. Pu stockpile quantities C2. Naval Pu reprocessing volume C3.
42 — Zheleznogorsk	C1–C2	Facility identity and Pu storage role C1. RT-2 completion status C2–C3. ADE-2 Pu quantities C2.
43 — NIIAR Dimitrovgrad	C1	Facility identity and reactor types C1. Cf-252 weapons application C1. BOR-60/Sarov data sharing C3 — ANALYTICALLY INFERRED.
44 — Kurchatov Institute	C1–C2	Facility identity and research role C1. Personnel exchange with Sarov C2–C3 — ANALYTICALLY INFERRED.

III. PRE-PUBLICATION CONFIDENCE GAPS

The following four assertions carry LOW CONFIDENCE — ANALYTICALLY INFERRED status and cannot be published as stated facts in a submission to SSRN, Georgetown SSP, or any peer-reviewed

venue. Each requires either corroboration from Tier 1 or credible Tier 2 sources, or explicit reframing as analytical inference with appropriate labeling.

1. DPRK lithium-6 procurement from NCCP (Node 33) — currently C3–C4. Requires a documented procurement chain from Tier 1 or credible Tier 2 sources, or warrants be reframed as 'analytically consistent with known DPRK procurement patterns' with explicit LOW CONFIDENCE — ANALYTICALLY INFERRED labeling.
2. Technology transfer from Sarov to DPRK (Node 38) — currently C4. Either support with credible open-source evidence (IISS, 38 North, NTI) or remove as an asserted fact and reframe as an analytical hypothesis requiring further investigation before publication.
3. BAM routing for anomalous nuclear shipments (Node 37) — currently C3. Appropriate as inference but warrants explicit labeling as LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED rather than stated as established practice.
4. BOR-60 weapons data sharing between NIIAR and Sarov (Node 43) — currently C3. Must be labeled LOW CONFIDENCE — ANALYTICALLY INFERRED based on institutional relationships, not documented data sharing.

IV. ACADEMIC LITERATURE ENGAGEMENT REQUIREMENTS

The following four assertions in the corpus are labeled ANALYTICALLY INFERRED and carry C3–C4 confidence. Per the CONF-01 framework, these assertions are documented as analytical hypotheses, not established facts. Each remains in the corpus with explicit confidence labeling and must not be cited as established findings in any submission.

ASSERTION	LOCATION	EXCLUSION NOTE
DPRK Li-6 procurement from NCCP	N33 NCCP profile; RST-01	C3–C4, ANALYTICALLY INFERRED. Do not cite as established DPRK procurement fact.
Technology transfer Sarov→DPRK	N38 Sarov profile	C4, ANALYTICALLY INFERRED. Do not cite as confirmed proliferation event.
BAM routing for anomalous nuclear shipments	N37 Tayshet; N45 Vanino	C3, ANALYTICALLY INFERRED. Do not cite as confirmed nuclear logistics fact.
BOR-60/ Sarov	N43 NIIAR profile	C3, ANALYTICALLY INFERRED. Do not cite as confirmed

data sharing		institutional relationship.

The March 2026 WP-2026 portfolio review identified academic literature integration as the primary gap for Georgetown SSP submission. The following literature categories warrant integration before SSRN or Georgetown SSP submission:

1. Nuclear fuel cycle technical baseline: IAEA Nuclear Energy Series publications on uranium enrichment, conversion, and fuel fabrication provide C1-quality technical grounding that should be explicitly cited in node profiles covering each process step.
2. Nonproliferation scholarship: Matthew Bunn and colleagues at the Harvard Belfer Center (particularly 'Preventing Nuclear Terrorism' series); Frank von Hippel at Princeton SPIA on enrichment and reprocessing constraints; NTI Global Nuclear Policy Program analytical reports on Russian fuel cycle security. These are the foundational academic references in this domain that submission reviewers will expect to see cited.
3. Sanctions effectiveness literature: Thomas Biersteker et al. (Graduate Institute Geneva) on targeted sanctions effectiveness; Clara Portela on EU sanctions design and compliance; Daniel Drezner (Tufts) on the theory of economic sanctions. The SENI scoring model's D1–D5 dimensions implicitly operationalize variables from this literature (legal authority, financial leverage, collateral cost, coalition requirement, time-to-impact) without citing the theoretical basis — a gap that Georgetown SSP reviewers will identify. The key engagement point is Drezner's sender-target cost-benefit framework as it applies to nuclear-specific sanctions.
4. Russian nuclear complex primary sources: Rosatom annual reports (available in English translation), TVEL JSC public filings, World Nuclear Association country profiles, and Russian-language sources including Atomnaya Energiya journal provide verifiable C1 sources. The IAEA Nuclear Energy Series (NES) on the Russian fuel cycle should be cited as the primary technical reference. These sources are cited at point of use in node profiles but not synthesized in a literature review section.

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CENTER FOR COMPETITIVE STATECRAFT AND STRATEGIC POLICY

WP-2026 SERIES

RECIPIENT STATE THREAT CHARACTERIZATION

Iran, DPRK, China — Demand-Side Program Assessment for SENI Enforcement Context

NOTICE *Independent policy research. All sources open-source. Confidence tiers per WP-2026-SENI-CONF-01. Does not constitute legal advice and does not establish classified findings.*

KEY JUDGMENTS

KJ 1

Iran's declared 60% U-235 enrichment and accumulated stockpile assessed as sufficient for multiple devices if further enriched to 90%+ represents the most acute near-term nonproliferation risk in the demand-side picture. Breakout timeline is assessed at weeks to months as of early 2026 [TIER 1 — IAEA Director General Reports 2024–2025].

LIKELY — MODERATE-HIGH CONFIDENCE

KJ 2

DPRK's operational nuclear weapons program and post-2022 Russia-DPRK strategic alignment create the most plausible pathway for Russian nuclear assistance — including centrifuge technology or lithium-6 — to reach a weapons program. This assessment does not establish that such transfer has occurred.

ROUGHLY EVEN ODDS — LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED

KJ 3

China's relationship with the Russian nuclear fuel cycle is primarily a legitimate commercial relationship. China as a demand-side state is primarily an enforcement constraint to manage, not a sanctions target. Supply substitution for Chinese VVER operators is the long-term structural resolution.

ALMOST CERTAINLY — HIGH CONFIDENCE

I. SCOPE AND LIMITATIONS

This document provides a policy-level characterization of the three principal recipient states for Russian nuclear material or technology transfer. It is not a node-by-node infrastructure map of foreign weapons programs. Its purpose is to define the demand-side context for SENI enforcement theory — establishing what each state's program needs, what it already has, and where Russian supply chain connections are analytically plausible. All assessments are drawn from published IAEA reports, congressional testimony, and open-source think tank analysis. Confidence tiers from WP-2026-SENI-CONF-01 apply throughout.

II. IRAN

A. Program Status

ATTRIBUTION	IAEA Director General reports to the Board of Governors document that Iran has enriched uranium to 60% U-235 at Fordow (Qom Province) and Natanz (Isfahan Province) under declared safeguards and has accumulated a stockpile of enriched material [TIER 1 — IAEA GOV/2025 series DG Reports]. Multiple government assessments, including ODNI's 2024 Annual Threat Assessment, indicate that Iran's breakout timeline is measured in weeks to months [TIER 1 — ODNI ATA 2024].
JUDGMENT	Official IAEA and ODNI reporting supports a moderate-high-confidence judgment that Iran's 60% enrichment stockpile is the most acute near-term nonproliferation risk in the demand-side picture. Iran has not been found to have produced weapons-grade material, but the technical gap between 60% and weapons-grade (90%+) is operationally bridgeable in a short breakout window.
LIMITS	The public record does not establish Iran's specific breakout timeline with precision. ODNI's 'weeks to months' assessment reflects a range of scenarios. The public record does not permit a firm conclusion on Iran's current weaponization work status.

B. Russian Supply Chain Connection

ATTRIBUTION	IAEA safeguards documentation identifies the Bushehr NPP as operating under declared IAEA safeguards with Russian fuel supply under TVEL JSC contracts [TIER 1 — IAEA CNPP Iran 2024]. State Department compliance reports have raised concerns about Russian-Iranian nuclear technical cooperation beyond Bushehr but have not established specific prohibited transfers [TIER 1 — State Department Compliance Reports 2022–2024].
JUDGMENT	Official reporting supports a moderate-confidence judgment that Russia's Bushehr fuel supply relationship creates institutional proximity and technical cooperation channels that extend beyond the declared civilian program. The proliferation concern is not Bushehr fuel itself but the knowledge transfer and relationship infrastructure built around it.
LIMITS	The public record does not establish specific instances of Russian centrifuge technology transfer to Iran. This assessment is LOW-MODERATE CONFIDENCE — ANALYTICALLY INFERRED based on Iranian centrifuge development trajectories and cannot serve as a published enforcement predicate without corroboration.

III. DPRK

A. Program Status

ATTRIBUTION	ODNI's 2024 Annual Threat Assessment assesses that North Korea has an

	operational nuclear weapons program with HEU enrichment capability at Yongbyon and potentially additional undeclared sites [TIER 1 — ODNI ATA 2024]. The DPRK has conducted six confirmed nuclear tests and is assessed to have deployed nuclear-armed ballistic missiles. IAEA monitoring was expelled in 2009 and has not resumed.
JUDGMENT	Official ODNI reporting supports a high-confidence judgment that DPRK has an operational nuclear weapons program — the most advanced and deployed of the three demand-side states assessed here. Its material bottleneck is not feedstock but enrichment capacity and weapons design refinement.
LIMITS	The public record does not establish the current size of DPRK's fissile material stockpile, the number of deployed warheads, or the operational status of any specific missile system. IAEA monitoring absence since 2009 creates a significant analytical gap.

B. Russian Supply Chain Connection

ATTRIBUTION	UN Panel of Experts reports to the UNSC 1718 Committee document post-2022 Russia-DPRK arms transfer relationships, including DPRK conventional munitions transfers to Russia [TIER 1 — UN Panel of Experts S/2024/215]. ODNI's 2024 ATA raises concerns about Russian military technology transfer to DPRK as reciprocation [TIER 1 — ODNI ATA 2024].
JUDGMENT	Official UN and ODNI reporting supports a low-moderate-confidence judgment that post-2022 Russia-DPRK strategic alignment creates conditions for nuclear-adjacent cooperation that did not exist before. This assessment does not establish that nuclear material or weapons design information has been transferred. The lithium-6 (Node 33) and centrifuge technology (Node 28) concerns are analytically plausible but LOW CONFIDENCE — ANALYTICALLY INFERRED.
LIMITS	The public record does not establish specific instances of nuclear material or weapons design technology transfer from Russia to DPRK post-2022. This assessment is LOW-MODERATE CONFIDENCE based on strategic alignment indicators, not documented nuclear transfers.

IV. CHINA

ATTRIBUTION	Pentagon's 2023 China Military Power Report assesses that China's nuclear warhead stockpile is expanding toward a 1,500-warhead goal by 2035 [TIER 1 — DoD China Military Power Report 2023]. IAEA safeguards documentation identifies CNNC as operating Chinese VVER-design reactors (Tianwan, Xudapu) with declared TVEL fuel supply [TIER 1 — IAEA CNPP China 2024].
JUDGMENT	Official reporting supports a high-confidence judgment that China's relationship with the Russian nuclear fuel cycle is a legitimate commercial relationship — TVEL fuel

	supply for declared civilian reactors under IAEA safeguards. China has its own enrichment capability and does not require Russian enrichment services for its weapons program. China is primarily an enforcement constraint, not a proliferation pathway.
LIMITS	The public record does not establish that Chinese VVER operators are seeking to reduce Russian fuel dependence. The public record does not establish what share of China's civilian reactor fuel requirements is currently met by TVEL versus domestic enrichment.

V. COMPARATIVE SUMMARY

	IRAN	DPRK	CHINA
Weapons status	Pre-nuclear; 60% HEU stockpile; weeks-to-months breakout [ODNI ATA 2024]	Operational; confirmed HEU; deployed missiles [ODNI ATA 2024]	Declared; expanding to ~1,500 warheads [DoD 2023]
Russian connection	Bushehr civilian + C2 institutional proximity	C2 arms transfer; C3 nuclear-specific — ANALYTICALLY INFERRED	C1 civilian fuel (TVEL); no weapons nexus
Primary SENI nodes	Rosatom-Iran tech cooperation entities	NCCP Li-6 (Node 33), UECC (Node 28), Vostochny (Node 36)	Krasnokamensk (Node 27), Angarsk (Node 31) — enforcement constraints
Primary instrument	JCPOA + EO 13382 Iran nexus	EO 13722 DPRK nexus	Supply substitution — not a sanctions target
Enforcement priority	High — breakout timeline acute	Highest — operational program; post-2022 alignment	N/A as sanctions target

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